BAYU AJI TRISASONGKO

ELECTRICAL ENGINEERING GRADUATE

Semarang, Indonesia | +62 822-2376-9735 | bayuajitrisasongko26@gmail.com | linkedin.com/in/bayuajitrisasongko | portobayu.vercel.app

PROFESSIONAL SUMMARY

Electrical Engineering graduate from Telkom University with strong multidisciplinary expertise across mechanical and electronics engineering. Experienced in designing and building a 5-DOF robotic manipulator arm for an undergraduate thesis, complemented by industrial internships in Mechanical Engineering and Electronics Process Engineering. Proficient in managing technical workflows from initial design and rapid prototyping to technical documentation, with a readiness to drive product development in dynamic engineering environments.

EDUCATION

Bachelor of Science in Electrical Engineering **September 2022 - August 2026**

Telkom University | Bandung, Indonesia

Vocational High School in Mechatronics Engineering **July 2018 - July 2022**

SMK Negeri 7 Semarang | Semarang, Indonesia

PROFESSIONAL EXPERIENCE

Process Engineering Intern **July 2025 - September 2025**

PT Promanufacture Indonesia (FORMULATRIX) | Salatiga, Indonesia

- Analyzed and supported Surface Mount Technology (SMT) line workflows, covering stencil printing, pick-and-place, reflow soldering, and optical inspection processes.
- Authored and updated Work Instructions (WI) for assembly and wiring procedures to establish standardized manufacturing guidelines for the production team.
- Designed PCB stencil panels using Easy-PC and exported Gerber files for vendor fabrication.
- Developed production-support brackets using Onshape CAD and fabricated functional parts utilizing 3D printing technologies.

Mechanical Engineering Intern **May 2021 - October 2021**

PT Promanufacture Indonesia (FORMULATRIX) | Semarang, Indonesia

- Modeled custom 3D components for laboratory automation equipment using Onshape in compliance with manufacturing requirements.
- Generated CNC toolpaths using OneCNC to prepare machining operations.
- Executed hands-on component fabrication processes, including 3D printing, deburring, filing, grinding, sheet metal bending, and mechanical assembly.
- Operated manual hand tools and power tools for precision assembly and component finishing.

PROJECT EXPERIENCE

Design of a Manipulator Arm for a Patient Service Robot in a Hospital **January 2026 - August 2026**

Telkom University | Bandung, Indonesia - Undergraduate Final Project

- Designed and fabricated a 5-DOF robotic manipulator arm utilizing 3D-printed PLA Cycloidal Gearboxes and an integrated closed-loop magnetic encoder control system, improving positioning accuracy by 79.8%.
- Validated a stable system payload capacity within the 250–1,000 g range, surpassing the initial 600 g target.
- Implemented Inverse Kinematics algorithms on a microcontroller to convert target Cartesian coordinates into precise joint angles.

Design of an Automatic Solar Panel Cooling System for Efficiency Enhancement **April 2025 - November 2025**

Telkom University | Bandung, Indonesia - Capstone Design

- Developed an automated temperature-controlled cooling system to maintain optimal operating temperatures for solar panels.
- Programmed automated control logic triggering a water pump when surface temperature exceeds 30°C and shutting off below 25°C.
- Achieved sensor measurement accuracy within ±0.6°C against a reference calibrated thermometer during subsystem testing.

Line Follower Robot Development **2024**

Independent Project | Bandung, Indonesia

- Built an autonomous line-tracking robot equipped with an array of 8 photodiode sensors and LEDs for track detection.
- Implemented motion control logic driven by real-time sensor inputs to achieve precise autonomous navigation.

Sumo Robot Development **2023**

Independent Project | Bandung, Indonesia

- Constructed a 1 kg, 20×20 cm competitive sumo robot using an acrylic chassis powered by 4 DC drive motors.
- Integrated a wireless Bluetooth control system, enabling seamless smartphone-based remote operation.
- Validated push force performance with load-pushing capability up to 4 kg.

ORGANIZATIONAL EXPERIENCE

Electronics and Intelligence Robotic Research Group (EIRRG) **2024 - 2025**

Mechanical Division, ROOBICS Team, Telkom University | Bandung, Indonesia

- Served as a core mechanical team member in designing and fabricating robot mechanical systems for the Kontes Robot ABU Indonesia (KRAI) 2025 national competition.

PKKMB Telkom University Committee (Freshmen Orientation) **2024**

Discipline and Motivation Team (DMT), Telkom University | Bandung, Indonesia

- Managed orientation protocols, student discipline, security, and crowd coordination for incoming freshmen throughout university orientation programs.

Keluarga Mahasiswa Teknik Elektro (KMTE) **2023 - 2026**

Mentor Division - QUOTE 2023, Telkom University | Bandung, Indonesia

- Mentored incoming Electrical Engineering freshmen by facilitating academic guidance, soft-skills development, and departmental orientation sessions.

COMPETITIONS

Kontes Robot ABU Indonesia (KRAI) **July 2025**

Contestant / Mechanical Team Member, ROOBICS Team, Telkom University | Bandung, Indonesia

- Engineered dribbling and shooting mechanisms for a basketball-playing robot, successfully advancing to the national stage at ITB Jatinangor.

SKILLS

Software: Fusion 360, Onshape, Easy-PC, EasyEDA, Bambu Studio, OneCNC, STM32CubeIDE, Arduino IDE

Mechanical: 3D CAD Modeling, Rapid Prototyping & 3D Printing, Manual Machining & Fabrication, Mechanical Assembly

Electrical & Embedded: PCB Design, Wiring & Harness Assembly, Sensor & Actuator Integration

Professional: Technical Documentation & Reporting, Cross-functional Collaboration, Problem Solving, Time Management

Languages: Indonesian (Native), English (EPrT 450, 2026)